

	Development and Production Operations Service CONTRACT for Sohrab Oil Field Wellpad Construction (3,4,8) Tender No.: SO-004-00082-03	
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SECTION 4

SCOPE OF WORK

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ARTICLE 1. INTRODUCTION

This Section describes the scope of the WORK for Well Pad Construction for well pads No 3,4 and 8 in Development and Production Operations Service Contract for Sohrab Oil Field Project which will be carried out based on the drawings and technical documents provided in Section 8 – TECHNICAL SPECIFICATIONS AND DRAWINGS.

ARTICLE 2. SCOPE OF WORK

The WORK shall comprise of three separate parts:

- a) Earthwork
 - Well pads
 - Burn pit access road
 - Burn pit
- b) Concrete and Masonry Works
 - Well pad
 - Burn pit
- c) Burning line
 - Burning line
 - Supports
- d) Other Works
 - Septic
 - Conductor Pipe
 - Steel Works

WORK is detailed below and shall include, but not limited to, the following major items:

2.1. Earthwork

2.1.1. Well pads

All layers of Well Pads area, shall be constructed based on "Typical Rough Grading Section for Well Pads" and Publication No 101 Second Revision: "Road General Technical Specification" for Earthworks in different layers and materials. Embankment slope shall be 1 vertical: 2 horizontals unless otherwise specified in drawings.

Main activities for Well Pad construction are as below:

- Leading embankment

Leading embankment shall be filled, graded and compacted with machinery up to 30 cm above the water level with suitable material according to Publication No 101 Second Revision: "Road General Technical Specification". The filling shall be compacted by machinery weight and final surface shall be compacted by roller.

Height of leading embankment layer is variable and will be specified based on existing water level. The soil for leading embankment shall be the mixture of local clay soil with fine sand.

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- Lime Soil Layer

Two 20cm layers shall be graded above leading embankment with suitable material and compacted up to 95% standard density at optimum moisture content as per the Modified Proctor Test to BS1377, AASHTO T180 or ASTM D1557.

The amount of lime shall be about 150 kg / m³. The exact amount of lime shall be determined by Soil Laboratory approved by COMPANY. All stabilizing activities shall be done in accordance with Publication No 268.

- Embankment

Embankment by local soil shall be done in accordance with related drawings and specifications. Required soil for compaction must be provided from local borrow pit. Bulk filling must be done layer by layer 15cm thickness. Each layer must be compacted up to 95% standard density at optimum moisture content as per the Modified Proctor test to BS1377, AASHTO T180 or ASTM D1557.

Height of embankment layer is variable and will be specified based on leading embankment height.

- Lime Soil Layer

A 15cm layer shall be graded above embankment (for gravel area) with suitable material and compacted up to 95% standard density at optimum moisture content as per the Modified Proctor Test to BS1377, AASHTO T180 or ASTM D1557.

The amount of lime shall be about 150 kg / m³. The exact amount of lime shall be determined by Soil Laboratory approved by COMPANY. All stabilizing activities shall be done in accordance with Publication No 268.

- Sub base Layer

Two 15cm sub base layers shall be executed on top of lime soil layer. The sub base shall be graded and compacted above lime soil layer up to 100% standard density at optimum moisture content as per the Modified Proctor test to BS1377, AASHTO T180 or ASTM D1557.

The sub base layer shall be in accordance with section 1-2-12 and the grade for sub base is grade B of Table 1-12 of Publication No. 101 Second Revision.

- Gravel

Finishing surface of well pads (except concrete areas) shall be covered by 7cm well graded gravel. The gravel shall be spreading in horizontal layer.

- Excavation

Excavation for cellar, rig foundation, pits and other area that needs to be excavated and shall be done in accordance with related documents and drawings. The excavated materials can be used for embankment subject to COMPANY approval unless otherwise transported to nominated disposal area. Excavation shall be executed in safety manner based on project specifications.

Excavation shall be done above and below the water level .The contractor should consider the conditions related to the drainage of water from the excavated site until the completion of the concrete works in his schedule and prices.

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Excavation shall be executed in safety manner based on HSE specifications and requirements as stipulated in Section 6 - HSE of CONTRACT.

- Bed Stabilizing with rubble stone

After excavation of waste pit, the bed shall be stabilized by 50 cm rubble stone.

2.1.2. Burn Pit Access Road

Burn pit access road shall be constructed with 6 meters width. Also, two pads (6*5) and (15*15) meters in access road shall be constructed based on access road cross section for green burner and related accessories. Embankment slope shall be 1 vertical: 2 horizontals unless otherwise specified in drawings. The Main activities for burn pit access road construction are as below:

- Leading embankment

Leading embankment shall be filled, graded and compacted with machinery up to water level with suitable material according to Publication No 101 Second Revision: "Road General Technical Specification" The filling shall be compacted by machinery weight and final surface shall be compacted by roller.

Height of leading embankment layer is variable and will be specified based on existing water level. The soil for leading embankment shall be the mixture of local clay soil with fine sand.

- Lime Soil Layer

Two 20cm layers shall be graded above leading embankment with suitable material and compacted up to 95% standard density at optimum moisture content as per the Modified Proctor Test to BS1377, AASHTO T180 or ASTM D1557.

The amount of lime shall be about 150 kg / m³. The exact amount of lime shall be determined by Soil Laboratory approved by COMPANY. All stabilizing activities shall be done in accordance with Publication No 268.

- Embankment

Embankment by local soil shall be done in accordance with related drawings and specifications. Required soil for compaction must be provided from local borrow pit. Bulk filling must be done layer by layer 15cm thickness. Each layer must be compacted up to 95% standard density at optimum moisture content as per the Modified Proctor test to BS1377, AASHTO T180 or ASTM D1557.

Height of embankment layer is variable and will be specified based on leading embankment height.

- Lime Soil Layer

A 15 cm layer shall be graded above leading embankment with suitable material and compacted up to 95% standard density at optimum moisture content as per the Modified Proctor Test to BS1377, AASHTO T180 or ASTM D1557.

The amount of lime shall be about 150 kg / m³. The exact amount of lime shall be determined by Soil Laboratory approved by COMPANY. All stabilizing activities shall be done in accordance with Publication No 268.

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- Sub base Layer

A 20cm sub base layer shall be executed on top of subgrade (soil + lime layer). The sub base shall be graded and compacted above lime soil layer up to 100% standard density at optimum moisture content as per the Modified Proctor test to BS1377, AASHTO T180 or ASTM D1557.

The sub base layer shall be in accordance with section 1-2-12 and the grade for sub base is grade B of Table 1-12 of Publication No. 101 Second Revision.

2.1.3. Burn Pit

Burn pits shall be constructed as per related drawings in Section 8. Embankment slope shall be 1 vertical: 2 horizontals unless otherwise specified in drawings. All earthworks (except sub-grade and sub-base) layers shall be executed as a ring with six-meter width on top of embankment. The Main activities for burn pit construction are as below:

- Leading embankment

Leading embankment shall be filled, graded and compacted with machinery up to water level with suitable material according to Publication No 101 Second Revision: "Road General Technical Specification" The filling shall be compacted by machinery weight and final surface shall be compacted by roller.

Height of leading embankment layer is variable and will be specified based on existing water level. The soil for leading embankment shall be the mixture of local clay soil with fine sand.

- Embankment

Embankment by local soil shall be done in accordance with related drawings and specifications. Required soil for compaction must be provided from local borrow pit. Bulk filling must be done layer by layer 15cm thickness. Each layer must be compacted up to 95% standard density at optimum moisture content as per the Modified Proctor test to BS1377, AASHTO T180 or ASTM D1557.

Height of embankment layer is variable and will be specified based on leading embankment height.

- Lime Soil Layer

A 15cm layer shall be graded above embankment with suitable material and compacted up to 95% standard density at optimum moisture content as per the Modified Proctor Test to BS1377, AASHTO T180 or ASTM D1557.

The amount of lime shall be about 150 kg / m³. The exact amount of lime shall be determined by Soil Laboratory approved by COMPANY. All stabilizing activities shall be done in accordance with Publication No 268.

- Excavation

Excavation for extra width of ring shall be done in accordance with related documents and drawings. The excavated materials can be used for embankment subject to COMPANY approval otherwise transported to nominated disposal area. Excavation shall be executed in safety manner based on project specifications.

For excavation under water level all technical and executive requirements shall be considered by Contractor.

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Excavation shall be executed in safety manner based on HSE specifications and requirements as stipulated in Section 6 - HSE of CONTRACT.

- Sub base Layer

A 20cm sub base layer shall be executed on top of subgrade (soil + lime layer). The sub base shall be graded and compacted above lime soil layer up to 100% standard density at optimum moisture content as per the Modified Proctor test to BS1377, AASHTO T180 or ASTM D1557.

The sub base layer shall be in accordance with section 1-2-12 and the sub base grade is grade B of Table 1-12 of Publication No. 101 Second Revision.

2.2. Concrete Works

2.2.1. Well pads construction

Based on related drawings, the masonry and concrete works shall be done as below:

- Rubble with Mortar

Under rig foundation, a masonry work includes rubble stone (سنگ لاشه) with mortar shall be executed. Minimum size of stone shall be 30cm and the stone work with mortar shall be executed by mason and in horizontal layer with proper mortar thickness.

The upper elevation of masonry work shall be prepared properly for reinforcement work of rig foundation.

- Reinforcement

Two type of steel rebar, reinforcing mesh and deformed steel bar are used for concrete work in Well Pad area.

The mesh type shall be AII and deformed steel bar shall be AIII.

All steel work shall be done in accordance with Section 8.

- Formwork

Formwork shall be executed with clean, new and proper steel formwork, accessories and fixtures in accordance with project specification.

Fixing of formworks to prevent concrete leakage or deforming of form work are CONTRACTOR's responsibility.

- Concrete Work

Three types of concrete, lean concrete, main concrete (C30), Main impermeable Concrete (30MPa), will be prepared and executed in accordance with related documents and drawings. Cement type for concrete shall be Portland type II based on geotechnical report. Mix design for two main concrete types shall be prepared by CONTRACTOR separately before starting the concrete work (Company approval is required). All concrete shall be prepared by batching plant.

Concrete parts of Well Pad are as follows:

- Rig Foundation

Main concrete (C30) shall be executed in accordance with related technical drawings (section 8). The tolerance level of the rig foundation shall be limited to 1cm for each one-meter grid.

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- Paving

Main concrete (C30) shall be executed in accordance with related technical drawings (section 8). The tolerance level of paving especially around the rig foundation shall be limited to 1.5cm.

- Concrete Blocks

Main concrete (C30) shall be executed in accordance with related technical drawings (section 8) for reinforced concrete blocks 2*1.2*0.5m. The blocks will be used for cutting and treatment area.

- Cellars

The concrete type for cellars is main impermeable Concrete (30 MPa). The cellar should be constructed as a sealed structure, therefore a water proof layer shall be prepared and executed by approved additive material and required water stop tape as related detail drawings (Section 8).

- Waste and Clear Pits

The concrete type for waste and clear pits is main impermeable Concrete (30 MPa). The pits should be constructed as a sealed structure, therefore a water proof layer shall be prepared and executed by approved additive material and required water stop tape as related detail drawings (Section 8).

- Coral

Main concrete (C30) for coral area shall be executed in accordance with related technical drawings (section 8).

- Treatment

Main concrete (C30) for treatment area shall be executed in accordance with related technical drawings (section 8).

- Interceptor (Oil Trap)

The concrete type for interceptor (oil trap) is main impermeable Concrete (30 MPa). The interceptor (oil trap) should be constructed as a sealed structure, therefore a water proof layer shall be prepared and executed by approved additive material and required water stop tape as related detail drawings (Section 8).

- Gas Oil Area

Main concrete (C30) for gas oil area shall be executed in accordance with related technical drawings (section 8).

- Ditches

At least three ditches shall be constructed in accordance with related drawings. Type of concrete for ditches is main concrete(C30).

Note: The embankment around the structures (back fill) should be done according to the technical specifications.

Note: Final surface of cellars, waste and clear pits and interceptor shall be sealed by using the special sealing materials.

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2.2.2. Burn Pit

Concrete works for burn pits include:

- Bed Stabilizing

Excavated bottom areas to be stabilized with 50cm rubble according to related drawings.

- Lean concrete of burn pit

10cm lean concrete shall be executed on the top of rubble layer as well as the second layer of Geotextile at the bottom of burn pit. A similar layer shall be executed on the walls after implementation of second layer of Geotextile as related technical drawings (Section 8)

- Geotextile

Two layers of geotextile(500gr/m²) shall be executed in accordance with related technical drawings (Section 8) for bottom and walls. At least 30cm overlapping shall be considered.

- Geomembrane

One layers of geomembrane(2mm) shall be executed in accordance with related technical drawings (Section 8) for bottom and walls. At least 5m roll width shall be prepared by CONTRACTOR. Welding of geomembrane shall be executed properly for avoiding any leakage in the burn pit.

- Reinforcement

The deformed steel bar is used for concrete work in Burn pit area. The deformed steel bar shall be AIII. All steel work shall be done in accordance with Section 8.

- Formwork

Formwork shall be executed with clean, new and proper steel formwork, accessories and fixtures in accordance with project specification.

Fixing of formworks (especially for walls) to prevent concrete leakage or deforming of form work are CONTRACTOR's responsibility.

- Main Concrete

The concrete type of burn pits is main impermeable Concrete (30 MPa). The burn pits should be constructed as a sealed structure, therefore a water proof layer shall be prepared and executed by approved additive material.

Note: Final surface of burn pit shall be sealed by using the special sealing materials.

2.3. Burning Line

2.3.1. Pipeline

- Preparing and executing a 320m (Approximate length) burning line with 4" pipe A106 grade B, sch STD.
- Preparing and executing a 12m casing with 6" pipe according to drawing.
- All pipeline tests shall be done in accordance with approved ITP.

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2.3.2. Pipe Supports

- Each 6m a support shall be constructed and installed in accordance with related drawings.
- All supports tests shall be done in accordance with approved ITP.

2.4. Other works

- Septic Tank

Two polyethylene septic tank each with the capacity of 25,000 liters shall be constructed in specified area in accordance with related drawing and procedure.

- Conductor Pipe Installation

42" conductor pipe inside the cellar shall be procured and prepared, fabricated in accordance with related drawings and installed properly in vertical position. The conductor pipe shall be driven by pile driving machine. The connection between conductor pipe and concrete shall be impermeable.

- Steel Work

In oil trap (interceptor) some steel work shall be done in accordance with related drawings.

- Slope Protection

The face layer of embankment in the direct contact of seasonal river or pool shall be protected by carcass stone with 30cm thickness and a layer of geotextile as related details in Section-8.

Note 1: All materials like geotextile, geomembrane, pipe line and pipe support from purchasing up to construction shall be done and tested based on project ITP with COMPANY approval.

Note 2: Concrete shall be prepared by batching plant with all technical requirement in accordance with Iranian Concrete Standards (ABA) and project ITP with COMPANY approval. Contractor shall provide 2 batching plant with the minimum total production volume of 400 m3

Note 3: All concrete (except lean concrete) shall be prepared by using Micro Silica. Concrete additive is preferred to be liquid or Gel.

Note 4: A water proof and sealing layers materials and procedures shall be proposed and provided by Contractor (Company approval is required).

The concrete mix design for impermeable concrete shall be prepared on the basis of impermeability by using Micro Silica and other additives. The required additives shall be provided from an approved vendor. Permeability control tests shall be done on the test sample before the operation. Water stop should be used in the connection of the floor concrete to the wall and construction joints.

ARTICLE 3. CONTRACTOR'S OBLIGATIONS

- 3.1. CONTRACTOR is responsible to check and endorse the BOQ, Specifications and Drawing as provided in Section 8 – TECHNICAL SPECIFICATIONS AND DRAWINGS, and specify any discrepancy between aforesaid documents at bid stage. No further claim in this regard is acceptable by COMPANY.

Note: The quantity of earth work is based on water level equal to 4.8 MSL that may change during the construction actual date. Therefore, the quantity of earth

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work will be changed in accordance with actual water level and will be based on mutual agreement of water level and new Civil 3D calculation.

- 3.2. CONTRACTOR is responsible for provision of the MATERIALS, construction water, electricity, and fuel for all facilities and machinery.
- 3.3. CONTRACTOR shall, expeditiously and in due time and as part of the WORK, obtain and maintain any and all such authorizations, permits and licenses as are necessary for the proper performance of the WORK and which are required to be obtained in the name of the CONTRACTOR. To the extent reasonable and necessary, the COMPANY shall assist the CONTRACTOR in this regard.
- 3.4. CONTRACTOR is responsible for protection and preservation of the services performed as well as of the MATERIALS, EQUIPMENT and goods at the WORKSITE from the COMMENCEMENT DATE to the completion of the services.
- 3.5. Within the technical safety and occupational health regulations and instructions, CONTRACTOR shall be liable for any loss and damages to third parties on the WORKSITE.
- 3.6. CONTRACTOR shall take all measures to prevent any loss or damage to the adjacent property and shall indemnify any damages to the neighboring estates, facilities and their finished products.
- 3.7. Other contractors may be engaged within the WORKSITE area. In such case CONTRACTOR shall exercise the necessary cooperation and coordination with such other contractors and shall not, with due respect to the coordinated overall time schedule, cause any impediment and restriction in rendering their services. CONTRACTOR also undertakes to avoid obstructing such Contractors and executives use of the non-private site facilities and not to disrupt and disturb their contractual performances.
- 3.8. In case for the illustration of any part of the WORK, preparation of any drawing is required, CONTRACTOR shall prepare such drawings free of any compensation.
- 3.9. CONTRACTOR shall carefully study all the drawings and specifications and informs the COMPANY in writing, of any errors giving reasons to resolve the matter and notify COMPANY accordingly.
- 3.10. CONTRACTOR shall, throughout the CONTRACT DURATION, prepare and have available at the WORKSITE as working documents, a complete set of As Built Drawings incorporating any and all changes implemented. These drawings shall be placed at the disposal of COMPANY for their study and review.
- 3.11. CONTRACTOR is responsible for providing the required machinery and EQUIPMENT. All machinery shall be on the good physical condition and suitable for the operation and shall be approved by the COMPANY.
- 3.12. CONTRACTOR shall follow the COMPANY's priorities and time schedule, in case of any unjustified delay, COMPANY shall have the rights to act in accordance with Articles 37 and or 38 of Section 2 - Terms and Conditions of CONTRACT.
- 3.13. Required concrete shall be provided by CONTRACTOR's batching plant and transporting to the WORKSITE based on the codes and standards.
- 3.14. Concrete mix design shall be reviewed and approved by COMPANY in advance.
- 3.15. All material shall be prepared in accordance with Procurement and QC procedures and approved ITP.

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- 3.16. All stone MATERIAL like granular, sub base and MATERIAL for concrete and etc. shall be obtained from the Andimeshk sources.
- 3.17. CONTRACTOR shall have a SUBCONTRACT with soil mechanic and technical laboratory for doing all required tests of operation in accordance with project ITP with COMPANY approval.
- 3.18. All kind of tests and inspections shall be applied during the operation based on the project specification and the COMPANY instructions.
- 3.19. The final dossier documentation which is to be provided by the CONTRACTOR shall contain the mechanical completion as-built dossier, red-line drawings, inspection, testing, certification and survey records for the facilities and all documentation of said dossier shall be prepared and submitted to COMPANY in three number of hard copies and two number of soft copies.
- 3.20. A 20 cm settlement has been considered in BOQ for leading embankment. More than this amount shall be considered by CONTRACTOR in his proposal.
- 3.21. Soil provision and responsibility
CONTRACTOR shall provide the appropriate soil and barrow pit(s) required for the implementation of the CONTRACT. CONTRACTOR is obliged to transport the soil from the barrow pit to the place of consumption which shall be considered in related Earth works rates (Table 2 - CONTRACT PRICE Breakdown of Section 3 – Schedule of Rates and Prices).
- 3.22. Mine and UXO Clearance
Before starting the embankment or excavation in the worksite or borrow pit area, CONTRACTOR shall consider the Mine and UXO clearance, and receive the approving clearance for working area from COMPANY's HSE and civil department.
- 3.23. CONTRACTOR shall be liable for execution of HSE rules and regulations specified in Section 6- HSE of the CONTRACT and COMPANY instruction. Contractor shall be ensured of observation of all HSE considerations by its personnel and shall abide to HSE rules at any time within the CONTRACT DURATION and shall cause its subcontractors of observing this matter.

ARTICLE 4. WBS

The following activities shall be observed by the CONTRACTOR.

1	Wellpad Construction 3
1.1	Earth Work
1.2	Concrete and Masonry Work
1.3	Burning Line & Support
1.4	Other Works
2	Wellpad Construction 8
2.1	Earth Work
2.2	Concrete and Masonary Work
2.3	Burning Line & Support
2.4	Other Works
3	Wellpad Construction 4

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3.1	Earth Work
3.2	Concrete and Masonary Work
3.3	Burning Line & Support
3.4	Other Works
4	Mobilization and Demobilization
4.1	Mobilization
4.2	Demobilization
5	HSE
	EMS (including the Clinic and related equipment and crew) and Ambulance Type B, including related equipment and crew
5.1	Fire Fighting Services and foamatic fire truck with one 16000-liter backup water tank including the crew and related equipment
5.2	